Derivation of a Clinical Prediction Rule for Pediatric Abusive Head Trauma*

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Objectives: Abusive head trauma is a leading cause of traumatic death and disability during infancy and early childhood. Evidence-based screening tools for abusive head trauma do not exist. Our research objectives were 1) to measure the predictive relationships between abusive head trauma and isolated, discriminating, and reliable clinical variables and 2) to derive a reliable, sensitive, abusive head trauma clinical prediction rule that—if validated—can inform pediatric intensivists' early decisions to launch (or forego) an evaluation for abuse.

Design: Prospective, multicenter, cross-sectional, observational.

Setting: Fourteen PICUs.

Patients: Acutely head-injured children less than 3 years old admitted for intensive care.

Interventions: None.

Measurements and Main Results: Applying a priori definitional criteria for abusive head trauma, we identified clinical variables that were discriminating and reliable, calculated likelihood ratios and post-test probabilities of abuse, and applied recursive partitioning to derive an abusive head trauma clinical prediction rule with maximum sensitivity—to help rule out abusive head trauma, if negative. Pretest probability (prevalence) of abusive head trauma in our study population was 0.45 (95 of 209). Post-test probabilities of abusive head trauma for isolated, discriminating, and reliable clinical variables ranged from 0.1 to 0.86. Some of these variables, when positive, shifted probability of abuse upward greatly but changed it little when negative. Other variables, when negative, largely excluded abusive head trauma but increased probability of abuse only slightly when positive. Some discriminating variables demonstrated

***See also p. 230**.

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The authors have not disclosed any potential conflicts of interest.

Dr. Hymel's research activities were supported, in part, by the Children's Hospital at Dartmouth.

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DOI: 10.1097/PCC.0b013e3182712b09

poor inter-rater reliability. A cluster of five discriminating and reliable variables available at or near the time of hospital admission identified 97% of study patients meeting a priori definitional criteria for abusive head trauma. Negative predictive value was 91%.

Conclusions: A more complete understanding of the specific predictive qualities of isolated, discriminating, and reliable variables could improve screening accuracy. If validated, a reliable, sensitive, abusive head trauma clinical prediction rule could be used by pediatric intensivists to calculate an evidence-based, patient-specific estimate of abuse probability that can inform—not dictate—their early decisions to launch (or forego) an evaluation for abuse. (*Pediatr Crit Care Med* 2013; 14:210–220)

Key Words: abusive head trauma; child abuse; decision rule; nonaccidental trauma; predictors; screening tools

A flawed decision *to forego* an abuse evaluation creates substantial risk of reinjury for a head-injured infant or young child returned to an abusive environment. In their 1999 study, Jenny et al (1) reported that 28% of acutely head-injured children whose abuse was initially missed or misdiagnosed were subsequently reinjured and 9% died during the period of diagnostic delay. A flawed decision *to launch* an abuse evaluation can also have negative consequences, including exacerbation of parental stress, compromise of the doctor-patient relationship, prolongation of the child's hospitalization, increased health care costs, exposure of the child to additional risks (radiation, sedation), and other unintended consequences (false-positive results, negative parental stigma). Unfortunately, multiple studies have shown that physicians struggle to define a reasonable medical certainty of abuse; demonstrate biases regarding child maltreatment related to race, socioeconomic status, and marital status; and demonstrate an inconsistent response to severe child maltreatment (1–12).

In the population of acutely head-injured infants and young children admitted for intensive care, both the prevalence and severity of abusive head trauma (AHT) are likely high. Should *every* acutely head-injured infant or young child admitted for intensive care be evaluated for abuse? Although this approach will minimize missed cases, we suspect that many pediatric intensivists defer abuse evaluations for some of their young, acutely head-injured patients when they deem it safe to do so. Faced with such challenging and important decisions, pediatric intensivists could benefit from evidence-based screening tools that can inform—not dictate—their initial estimates of abuse probability. To date, few investigators or clinicians have proposed evidence-based approaches for child physical abuse screening or diagnosis (13–19).

The language and logic of evidence-based screening and diagnosis is well-summarized in the *Users' Guide to the Medical Literature* (20). Like sensitivity and specificity, likelihood ratios describe the discriminatory power of clinical findings. The positive likelihood ratio tells us how much to increase (or decrease) the probability of a diagnosis if the clinical finding is present. The negative likelihood ratio tells us how much to decrease (or increase) the probability if the clinical finding is absent. The advantage of likelihood ratios is how simply they can be used to estimate the post-test probability of a diagnosis (**Table 1**). In general, likelihood ratios greater than 10 predict large and often conclusive increases in the probability of a diagnosis and likelihood ratios less than 0.1 predict large and often conclusive decreases in the probability.

A clinical prediction rule is a tool that measures and then combines the predictive contributions of multiple clinical findings or tests to estimate or predict the probability of a diagnosis, prognosis, or response to therapy *for an individual patient*. Clinical prediction rules are most likely to be useful when clinical decision making is complex, when clinical stakes are high, or when there are opportunities to achieve cost savings without compromising patient care (20, 21). High-performance clinical prediction rules developed as screening tools to help exclude a diagnosis demonstrate sensitivity greater than 0.95 and negative likelihood ratio less than 0.1 within narrow confidence limits (22). By established convention, development of a clinical prediction rule proceeds in three sequential phases: derivation, validation, and implementation (20, 21). Each phase requires a multicenter study. In this article, we describe the results of a multicenter derivation study intended ultimately to facilitate screening for AHT at or near the time of admission for intensive care.

Focusing exclusively on a population of acutely head-injured infants and young children admitted for intensive care, our first research objective was to measure the predictive relationships between AHT and isolated, discriminating, and reliable clinical variables. Our second objective was to derive an AHT clinical prediction rule 1) that is simple and reliable; 2) that could be applied at or near the time of PICU admission; 3) that is highly sensitive—to help clinicians exclude abuse, if negative; and 4) that *if* subsequently validated, could facilitate calculation of an evidence-based, patient-specific estimate of the probability of abuse that can inform clinicians' early decisions to launch (or forego) an evaluation for abuse. We did *not* seek to derive an AHT clinical prediction rule with high specificity, to confirm abuse, if positive.

MATERIALS AND METHODS

Patients and Settings

Between February 2010 and August 2011, we conducted a prospective study of acutely head-injured children less than 3 years old hospitalized in 14 PICUs.

Study Design

This was an observational, cross-sectional study designed to measure the predictive relationships between clinical variables and AHT at the time of acute hospitalization for intensive care. Each participating site received approval from the Human Subjects Research Committee with a waiver of informed consent.

Inclusion and Exclusion Criteria

Eligible patients were children younger than 3 years old hospitalized acutely in a PICU for treatment of symptomatic, acute, closed

TABLE 1. Evidence-Based Diagnosis

	Condition Present	Condition Absent	Totals
Finding present	A	B	A + B
Finding absent	C	D	C + D
Totals	A + C	B + D	A + B + C + D
To calculate post-test probabilities, apply Bayes' theorem			
SENS	Sensitivity		A/(A + C)
SPEC	Specificity		D/(B + D)
LR(+)	Positive likelihood ratio		SENS/(1 − SPEC)
LR(−)	Negative likelihood ratio		(1 − SENS)/SPEC
PREV	Prevalence (pretest probability)		(A + C)/(A + B + C + D)
SPTO	Study pretest odds		PREV/(1 − PREV)
PPTO	Positive post-test odds		SPTO × LR(+)
NPTO	Negative post-test odds		SPTO × LR(−)
PPTP	Positive post-test probability		PPTO/(PPTO + 1)
NPTP	Negative post-test probability		NPTO/(NPTO + 1)

(nonpenetrating), traumatic, cranial or intracranial injuries confirmed by computed tomography or magnetic resonance imaging. Patients were excluded if initial neuroimaging 1) revealed clear evidence of preexisting brain malformation, disease, infection, or hypoxia-ischemia or 2) failed to reveal any evidence of acute head trauma. Because they present no real diagnostic confusion with AHT, patients with head injuries resulting from collisions involving motor vehicles were also excluded.

Data Collection and Management

At every participating site, research coordinators screened PICU admission logs to identify patients meeting inclusion criteria. Participating sites were required by protocol to enroll over 90% of their eligible subjects based on monthly audits. Local investigators captured data in access-controlled, online data registries, using standardized electronic data forms that had been piloted extensively at Dartmouth-Hitchcock Medical Center.

Standard-of-care treatments yielded all required data. Subject screening, enrollment, and demographic data were captured by research coordinators. Data regarding potential historical, clinical, or radiological predictors of AHT were captured by pediatric intensivists or treating providers prior to hospital discharge. To minimize observational bias, historical data and data regarding child abuse diagnostic testing, results, final diagnostic impressions, and response were captured independently, at the time of hospital discharge, by child abuse specialists (when consulted) or by treating medical providers (when necessary). Missing or pending data were captured retrospectively after hospital discharge.

Data integrity and internal consistency were assessed by the study principal investigator (K.P.H.) as soon as possible after complete entry of patient-specific data in the primary data registry. Concerns about data integrity or internal consistency prompted queries to local investigators that were tracked until resolution. To assess inter-rater reliability, investigators at every site captured blinded, duplicate data on 20% of their study patients and entered these data into an online data registry identical to our primary data registry. Analyses were undertaken using Excel (Microsoft, Redman, WA) and R 2.3.1 (R Development Core Team, Vienna, Austria).

Outcome Measure

Gold standard definitional criteria for AHT do not exist. In their absence, AHT investigators must utilize a priori study definitional criteria for AHT specifically designed to minimize circular reasoning. The a priori definitional criteria for AHT used in this study (**Table 2**) reflect the results of a focused effort to define AHT and mirror or replicate definitional criteria used in other peer-reviewed studies (23–26). To minimize circular reasoning, they contain no references to specific intracranial injuries, to injury severity, or to any of the historical, clinical, or radiological variables considered for inclusion in our AHT clinical prediction rule. Patients meeting any one or more of our a priori study definitional criteria for AHT were categorized as abused. All remaining patients were categorized as other.

To broadly assess their diagnostic accuracy, we compared patient sorting based on these study definitional criteria to the final, consensus, diagnostic impressions of treating (and consulting) physicians at the time of hospital discharge—when all subsequent evaluations for abuse (if ordered) were complete. For this secondary analysis, treating physicians' final diagnostic impressions of definitive or probable AHT were collapsed into a single-abused category, and their final diagnostic impressions of undetermined, probable, or definitive non-abusive head trauma were collapsed into a single other category.

Assessment of Isolated Candidate Predictor Variables

To measure the predictive relationships between AHT and isolated, discriminating, and reliable clinical variables, we 1) captured dichotomous data regarding 45 historical, clinical, or radiological variables; 2) applied bivariate (chi-square) analysis to identify variables that discriminated significantly between

TABLE 2. A Priori Study Definitional Criteria for Pediatric Abusive Head Trauma

A Study Patient Was Categorized As Abused If…	Number of Patients Meeting Criterion[a]
The primary caregiver[b] admitted abusive acts	14
Abusive acts by the primary caregiver[b] were witnessed by an unbiased, independent observer	1
The primary caregiver[b] specifically denied that the preambulatory child in his/her care had experienced any head trauma	32
The primary caregiver[b] provided an account of the child's head injury event that was clearly historically inconsistent with repetition over time	19
The primary caregiver[b] provided an account of the child's head injury event that was clearly developmentally inconsistent with the child's known (or expected) gross motor skills	12
Further workup confirmed the presence of two or more categories of extracranial injuries considered moderately or highly suspicious for abuse[c]	53

[a]Ninety-five (45%) of 209 study patients met one or more study definitional criteria for pediatric abusive head trauma. Some patients met multiple definitional criteria.

[b]Defined a priori as the person responsible for the child when he/she was acutely head injured or first became clearly and persistently ill with clinical signs linked to acute traumatic head injuries confirmed by computed tomography or magnetic resonance imaging.

[c]Including classic metaphyseal lesion fracture(s) or epiphyseal separation(s); rib fracture(s); fracture(s) of the scapula or sternum; fracture(s) of digits; vertebral body fracture(s), dislocation(s), or fracture(s) of spinous process(es); skin bruising, abrasion(s), or laceration(s) in two or more distinct locations other than knees, shins, or elbows; patterned skin bruising or dry contact burn(s); scalding burn(s) with uniform depth, clear lines of demarcation, and paucity of splash marks; confirmed intra-abdominal injuries; retinoschisis confirmed by an ophthalmologist; retinal hemorrhages described by an ophthalmologist as dense, extensive, covering a large surface area and/or extending to the ora serrata.

abused vs. other study patients; 3) calculated each variable's inter-rater reliability (kappa, κ); 4) identified variables that were both discriminating ($p \leq 0.001$ by chi-square testing, applying Bonferroni's adjustment to minimize risk of type I error) and reliable (lower limit of 95% confidence interval for $\kappa > 0.4$); 5) calculated the overall prevalence (pretest probability) of AHT in our PICU study patient population; and 6) calculated the specific prevalence, sensitivity, specificity, likelihood ratios, and post-test probabilities of abuse for every discriminating and reliable variable.

Clinical Prediction Rule Derivation

Every discriminating and reliable variable available at or near the time of PICU admission became a candidate for inclusion in our AHT clinical prediction rule. We derived our rule by fitting a classification tree via binary recursive partitioning. The partitioning algorithm works by identifying splitting variables that maximize the homogeneity of patients with respect to abuse status (abused vs. other). Thus, a parent node is split into two child nodes, and each child node is classified as either abused or other so that the diagnostic accuracy of each child node exceeds that of the parent node. The algorithm continues as each child node is either split by another predictor variable or identified as a terminal node with terminal classification. The final classification tree is inclusive and exhaustive, so that all patients can be evaluated under each splitting criterion and a terminal classification is applied to each patient. Standard methods for fitting classification trees would equally penalize missed cases of AHT and other head trauma. In order to minimize the number of AHT cases missed, we applied a weighted loss function in which the penalty for missed cases of AHT was three times that of the penalty for missed cases of other head trauma. In order to minimize model complexity and avoid model overfitting, the partitioning algorithm only split nodes in the tree with 20 or more patients (any node with fewer than 20 patients was fixed as a terminal node).

Assessment of Clinical Prediction Rule Classification Performance and Screening Accuracy

To assess the classification performance and screening accuracy of our derived AHT clinical prediction rule, we 1)

TABLE 3. Demographics ($n = 209$)

Age at the Time of Hospital Admission (mos)	
Median	7
Mean	10.5
SD (range)	9.8 (0–35)
Gender, *n* (%)	
Male	132 (63)
Race, *n* (%)	
White or white-Hispanic	174 (83)
Black, African American, or Black-Hispanic	19 (9)
Other	16 (8)
Ethnicity, *n* (%)	
Hispanic or Latino	61 (29)
Not Hispanic or Latino	144 (69)
Ethnicity unknown	4 (2)

applied our derived rule to every child in our study patient population; 2) categorized study patients who manifested any one or more of the variables included in the derived rule as higher risk; 3) categorized study patients who manifested none of the included variables as lower risk; 4) created a 2 × 2 contingency table; 5) compared patient sorting based on our derived AHT clinical prediction rule (higher risk vs. lower risk) to that based on the a priori study definitional criteria for AHT (abused vs. other); and 6) calculated the derived clinical prediction rule's sensitivity, specificity, predictive values, and likelihood ratios.

RESULTS

Investigators at 14 participating sites captured complete data regarding 209 (99.5%) of 210 eligible PICU patients identified through local screening of admissions. Their demographic characteristics are summarized in **Table 3**. Questions regarding data integrity or internal consistency prompted 26 data queries from the study PI (Dr. Hymel) to local investigators. All were resolved at the discretion of local investigators. To test inter-rater reliability, duplicate data were captured blindly regarding 42 (20%) of 209 children.

Ninety-five (45%) of 209 children in our study patient population met one or more of the a priori study definitional criteria for AHT, including 14 whose perpetrators admitted abuse and 53 who manifested two or more categories of extracranial injuries considered moderately or highly suspicious for abuse (Table 2). The remaining 114 (55%) of 209 children were categorized as other.

Twenty-two (49%) of 45 variables discriminated significantly ($p \leq 0.001$ by chi-square testing, applying Bonferroni's adjustment to minimize risk of type 1 error) among abused vs. other study patients. Thirty-two (71%) of 45 variables demonstrated acceptable inter-rater reliability (lower limit of 95% confidence interval for $\kappa > 0.4$). Twenty (44%) of 45 were both discriminating and reliable (**Table 4**), including 13 variables available at or near the time of PICU admission. Likelihood ratios for the 20 (early or late) discriminating and reliable variables ranged from 0.13 to 7.2. Applying Bayes' theorem (Table 1), post-test probabilities of AHT for these 20 variables ranged from 0.1 to 0.86 (**Table 5** and **Fig. 1**).

Binary recursive partitioning identified a cluster of five discriminating and reliable variables available

TABLE 4. Results of Chi-Square and Inter-Rater Reliability[a] Testing of 45 Candidate Predictor Variables

Early Variables or Findings	*p*	κ (95% Confidence Interval)
Acute encephalopathy[b] PTA	< 0.001	0.95 (0.86, 1.00)
Acute encephalopathy[b], absent or resolved PTA	< 0.001	0.90 (0.76, 1.00)
Acute respiratory compromise[c] PTA	< 0.001	0.90 (0.77, 1.00)
Acute circulatory compromise[d] PTA	< 0.001	0.58 (0.26, 0.89)
Seizure(s) PTA	< 0.001	0.78 (0.57, 0.98)
Seizure(s) or acute encephalopathy[b] PTA	< 0.001	0.95 (0.86, 1.00)
Any craniofacial soft tissue injuries[e]	0.349	0.75 (0.53, 0.98)
Any bruising in a child < 4 months old	0.082	0.87 (0.63, 1.00)
Any bruising in a child not yet cruising	0.006	0.49 (0.11, 0.86)
Bruising involving the child's ear(s), neck, or torso[f]	< 0.001	0.88 (0.72, 1.00)
Initial hematocrit (%) < 28	0.126	0.69 (0.39, 0.98)
Initial partial thromboplastin-international normalized ratio > 1	0.163	0.79 (0.60, 0.99)
Any skull fracture(s)	< 0.001	0.90 (0.77, 1.00)
An isolated, unilateral, nondiastatic, linear, parietal skull fracture	0.001	0.69 (0.41, 0.98)
Any complex skull fracture(s)[g]	0.034	0.72 (0.48, 0.95)
Any skull fracture(s) other than an isolated, unilateral, nondiastatic, linear, parietal skull fracture	< 0.001	0.80 (0.62, 0.99)
Epidural hemorrhage(s)	0.062	0.71 (0.40, 1.00)
Any subdural hemorrhage(s) or fluid collection(s)	< 0.001	0.90 (0.76, 1.00)
Single density, hyperdense, subdural hemorrhage(s), or fluid collection(s)[h]	0.670	0.66 (0.43, 0.89)
Single density, hypodense, subdural hemorrhage(s), or fluid collection(s)[h]	0.110	0.36 (0.00, 1.00)
Mixed density subdural hemorrhage(s) or fluid collection(s)[h]	< 0.001	0.38 (0.00, 0.89)
Isolated, small, focal, subdural hemorrhage(s)	0.150	0.88 (0.64, 1.00)
Interhemispheric subdural hemorrhage	< 0.001	0.68 (0.44, 0.92)
Unilateral subdural hemorrhage(s)	0.018	0.89 (0.74, 1.00)
Bilateral subdural hemorrhage(s) or fluid collection(s)	< 0.001	0.75 (0.53, 0.98)

(*Continued*)

TABLE 4. (*Continued*) Results of Chi-Square and Inter-Rater Reliability[a] Testing of 45 Candidate Predictor Variables

Early Variables or Findings	*p*	κ (95% Confidence Interval)
Interhemispheric or bilateral subdural hemorrhage(s)	<0.001	0.80 (0.61, 0.99)
Any subarachnoid hemorrhage(s)	0.096	0.69 (0.46, 0.92)
Any brain parenchymal contusion(s), laceration(s), or hemorrhage(s)	0.126	0.68 (0.42, 0.94)
Brain parenchymal contusion(s), laceration(s), or hemorrhage(s) limited to the cortical brain	0.088	0.53 (0.02, 1.00)
Brain parenchymal contusion(s), laceration(s), or hemorrhage(s) involving the subcortical brain	0.174	0.81 (0.54, 1.00)
Brain parenchymal contusion(s), laceration(s), or hemorrhage(s) compatible with diffuse axonal injury[h,i]	0.062	0.69 (0.34, 1.00)
Late Variables or Findings	***p***	**κ [95% Confidence Interval]**
Acute encephalopathy[b] PTA, resolved within 24 hrs	0.968	0.79 (0.38, 1.00)
Acute encephalopathy[b] PTA, lasting > 24 hrs	< 0.001	0.94 (0.83, 1.00)
Acute encephalopathy[b] PTA, lasting > 24 hrs, without flaccidity, decorticate, or decerebrate posturing	0.070	1.00 (1.00, 1.00)
Acute encephalopathy[b] PTA lasting > 24 hrs, with flaccidity, decorticate, or decerebrate posturing	< 0.001	0.93 (0.80, 1.00)
Delayed impairment or loss of consciousness with onset after hospital admission	0.047	0.36 (0.00, 1.00)
AST or ALT level > 80 IU/L any time after hospital admission, if completed	0.007	0.83 (0.65, 1.00)
Any brain hypoxia–ischemia or swelling	< 0.001	0.84 (0.66, 1.00)
Brain hypoxia–ischemia or swelling limited to the cortical brain	0.018	0.29 (0.00, 0.95)
Brain hypoxia–ischemia or swelling involving the subcortical (or deeper) brain	< 0.001	0.81 (0.60, 1.00)
Unilateral brain hypoxia–ischemia or swelling	0.163	NC
Bilateral brain hypoxia–ischemia or swelling	< 0.001	0.88 (0.73, 1.00)
Any brain herniation or shifting	0.508	0.72 (0.45, 0.98)
Scalp, skull, or epidural "greatest depth" of visible (1° or 2°) injury	< 0.001	0.78 (0.54, 1.00)
Subcortical (or deeper) brain "greatest depth" of visible (1° or 2°) injury	< 0.001	0.75 (0.53, 0.98)

p = probability value resulting from chi-square analysis; PTA = prior to admission, NC = not calculable; AST = aspartate aminotransferase; ALT = alanine aminotransferase.

[a]Based on comparisons of duplicate data captured blindly regarding 42 (20%) of 209 study patients.

[b]Defined as any loss or alteration of consciousness.

[c]Including infrequent or labored respirations, apnea, or any requirement for intubation or assisted ventilation.

[d]Including bradycardia, hypotension, delayed capillary refill, cardiac arrest, or any requirement for urgent volume expansion, chest compressions, or vasoactive therapy.

[e]Including bruising, abrasion(s), laceration(s), or swelling; subgaleal hematoma(s) or cephalohematoma(s).

[f]Including the chest, abdomen, genitourinary region, back, and buttocks.

[g]Including skull fractures that are multiple, bilateral, branching or comminuted, diastatic, stellate, or crossing suture lines.

[h]If CT completed.

[i]Defined as CT evidence of multiple, small, parenchymal, or intraventricular hemorrhages.

at or near the time of PICU admission that—alone or in combination—identified 92 (97%) of 95 children in our study patient population who met one or more of the a priori study definitional criteria for abuse (**Fig. 2**). These variables included 1) acute respiratory compromise prior to admission; 2) seizure(s) or acute encephalopathy prior to admission; 3) bruising of the ear(s), neck, or torso; 4) interhemispheric or bilateral subdural hemorrhage(s) or fluid collection(s); and 5) any skull fracture(s) other than an isolated, unilateral, nondiastatic, linear, and parietal skull fracture. Specificity, positive predictive value, negative predictive value, and negative likelihood ratio for the AHT clinical prediction rule incorporating these five variables were 0.27, 0.53, 0.91, and 0.12, respectively (**Table 6**). Other, smaller combinations of these variables also demonstrated sensitivities for AHT exceeding 90% (**Fig. 3**).

Only 152 (73%) of 209 study subjects underwent skeletal surveys. One hundred and forty (67%) of 209 underwent ophthalmology evaluations. All study subjects sorted as lower risk by our derived (five-variable) AHT clinical prediction rule (n = 34) who underwent skeletal survey (n = 23) or ophthalmology evaluations (n = 16) had negative results.

Secondary analysis demonstrated no statistically significant difference (two-tail McNemar's test, 0.152) among patient sorting based on a priori study definitional criteria for AHT vs. that based on

TABLE 5. Predictive Qualities of 20 Isolated, Discriminating,[a] and Reliable[b] Variables

Early Variables or Findings	SENS	SPEC	LR(+) [95% CI]	LR(–) [95% CI]	PPTP [95% CI]	NPTP [95% CI]	PREV
Acute encephalopathy[c] PTA	0.73	0.56	1.66 [1.30, 2.11]	0.49 [0.34, 0.70]	0.58 [0.52, 0.64]	0.29 [0.22, 0.37]	0.57
Acute encephalopathy[c], absent or resolved PTA	0.45	0.28	0.63 [0.49, 0.81]	1.95 [1.38, 2.76]	0.34 [0.29, 0.40]	0.62 [0.53, 0.70]	0.60
Acute respiratory compromise[d] PTA	0.59	0.73	2.17 [1.54, 3.06]	0.56 [0.43, 0.74]	0.64 [0.56, 0.72]	0.32 [0.26, 0.38]	0.42
Seizure(s) PTA	0.48	0.83	2.91 [1.83, 4.60]	0.62 [0.50, 0.76]	0.71 [0.60, 0.79]	0.34 [0.29, 0.39]	0.31
Seizure(s) or acute encephalopathy[c] PTA	0.78	0.55	1.74 [1.38, 2.19]	0.40 [0.26, 0.60]	0.59 [0.53, 0.65]	0.25 [0.18, 0.33]	0.60
Bruising involving the child's ear(s), neck, or torso[e]	0.44	0.94	7.20 [3.39, 15]	0.59 [0.49, 0.72]	0.86 [0.74, 0.93]	0.33 [0.29, 0.37]	0.23
Any skull fracture(s)	0.24	0.34	0.37 [0.25, 0.54]	2.22 [1.68, 2.93]	0.24 [0.17, 0.31]	0.65 [0.58, 0.71]	0.47
An isolated, unilateral, nondiastatic, linear, parietal skull fracture	0.07	0.76	0.31 [0.14, 0.68]	1.21 [1.08, 1.36]	0.21 [0.10, 0.36]	0.50 [0.47, 0.53]	0.16
Any skull fracture(s) other than an isolated, unilateral, nondiastatic, linear, parietal skull fracture	0.17	0.58	0.40 [0.24, 0.66]	1.44 [1.20, 1.72]	0.25 [0.17, 0.35]	0.55 [0.50, 0.59]	0.31
Any subdural hemorrhage(s) or fluid collection(s)	0.90	0.47	1.67 [1.39, 2.01]	0.23 [0.12, 0.42]	0.58 [0.54, 0.63]	0.16 [0.09, 0.26]	0.70
Interhemispheric subdural hemorrhage	0.57	0.82	3.09 [2.02, 4.72]	0.53 [0.41, 0.68]	0.72 [0.63, 0.80]	0.31 [0.25, 0.36]	0.36
Bilateral subdural hemorrhage(s) or fluid collection(s)	0.57	0.84	3.60 [2.28, 5.70]	0.51 [0.40, 0.65]	0.75 [0.66, 0.83]	0.30 [0.25, 0.35]	0.34
Interhemispheric or bilateral subdural hemorrhage(s) or fluid collection(s)	0.73	0.75	2.96 [2.10, 4.17]	0.36 [0.26, 0.51]	0.71 [0.64, 0.78]	0.23 [0.18, 0.30]	0.46
Late Variables or Findings	**SENS**	**SPEC**	**LR(+) [95% CI]**	**LR(–) [95% CI]**	**PPTP [95% CI]**	**NPTP [95% CI]**	**PREV**
Acute encephalopathy[c] PTA, lasting > 24 hrs	0.43	0.83	2.59 [1.62, 4.15]	0.68 [0.56, 0.83]	0.68 [0.57, 0.78]	0.36 [0.32, 0.41]	0.29
Acute encephalopathy[c] PTA, lasting >24 hrs, with flaccidity, decorticate or decerebrate posturing	0.28	0.90	2.95 [1.54, 5.62]	0.79 [0.69, 0.91]	0.71 [0.56, 0.82]	0.40 [0.37, 0.43]	0.18
Any brain hypoxia-ischemia or swelling	0.51	0.81	2.62 [1.71, 4.00]	0.61 [0.49, 0.77]	0.69 [0.59, 0.77]	0.34 [0.29, 0.39]	0.34
Brain hypoxia-ischemia or swelling involving the subcortical (or deeper) brain	0.32	0.89	2.77 [1.53, 5.00]	0.77 [0.66, 0.90]	0.70 [0.56, 0.81]	0.39 [0.35, 0.43]	0.21
Bilateral brain hypoxia-ischemia or swelling	0.39	0.87	2.96 [1.73, 5.05]	0.70 [0.59, 0.84]	0.71 [0.59, 0.81]	0.37 [0.33, 0.41]	0.25
Scalp, skull, or epidural "greatest depth" of visible (1° or 2°) injury	0.04	0.68	0.13 [0.05, 0.36]	1.4 [1.23, 1.60]	0.1 [0.04, 0.23]	0.54 [0.51, 0.57]	0.19
Subcortical (or deeper) brain "greatest depth" of visible (1° or 2°) injury	0.36	0.85	2.4 [1.43, 4.02]	0.75 [0.64, 0.89]	0.67 [0.54, 0.77]	0.39 [0.35, 0.43]	0.24

SENS = sensitivity; SPEC = specificity; LR (+) = positive likelihood ratio; CI = confidence interval; LR (–) = negative likelihood ratio; PPTP = positive post-test probability; NPTP = negative post-test probability; PREV = prevalence; PTA = prior to admission.

[a] $p \leq 0.001$ by chi-square testing, applying Bonferroni's adjustment to minimize risk of type 1 error.

[b] Lower limit of 95% CI for $\kappa > 0.4$, based on comparisons of duplicate data captured blindly regarding 42 (20%) of 209 study subjects.

[c] Defined as any loss or alteration of consciousness.

[d] Including infrequent or labored respirations, apnea, or any requirement for intubation or assisted ventilation.

[e] Including the chest, abdomen, genitourinary region, back, and buttocks.

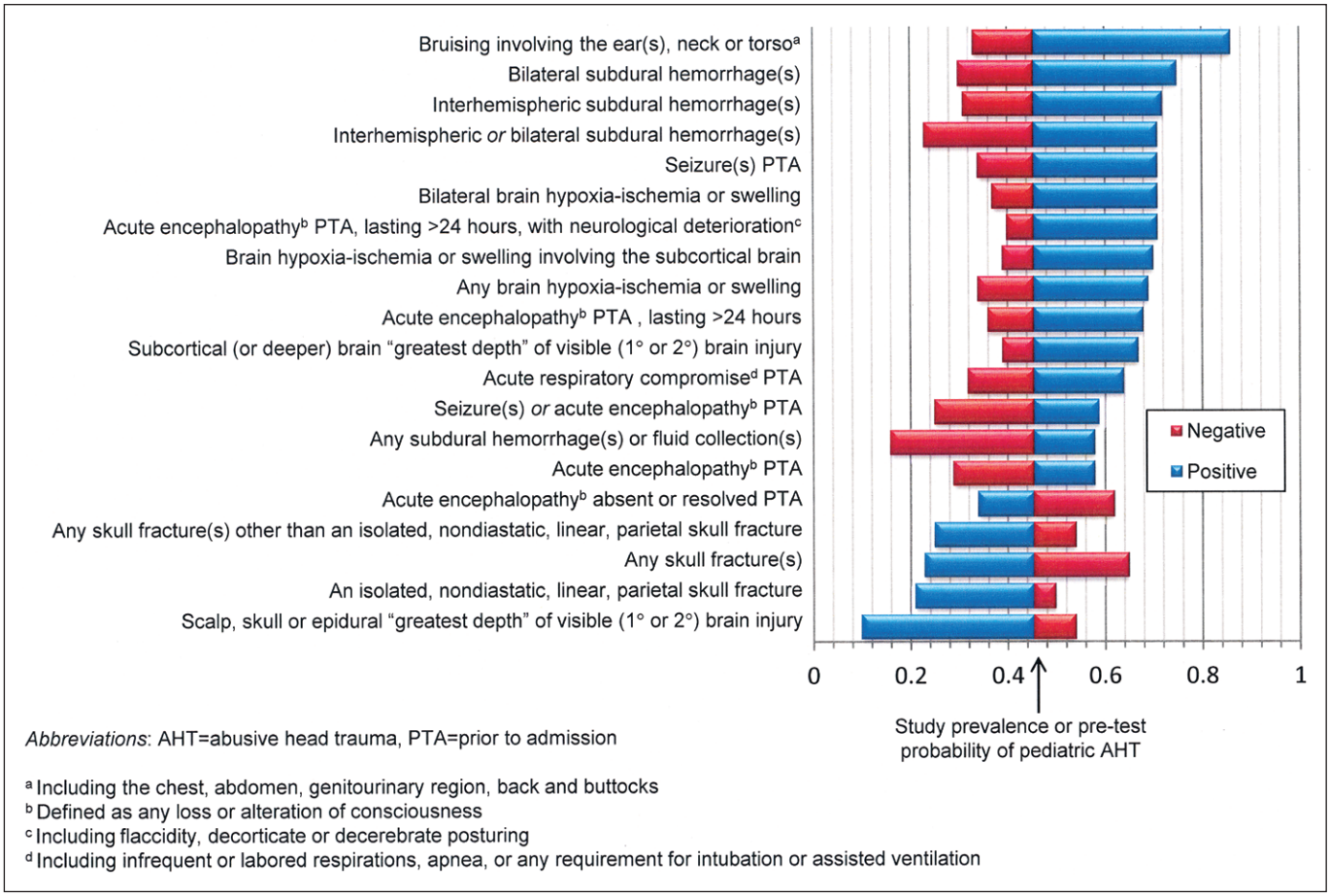

Figure 1. Estimated post-test probabilities of AHT for isolated, discriminating, and reliable variables.

treating (and consulting) physicians' final, consensus, diagnostic impressions at the time of hospital discharge, when all subsequent evaluations for abuse (if ordered) were complete (**Table 7**). Two of three "missed" patients (i.e., study patients sorted as *lower risk* by our derived [five-variable] AHT clinical prediction rule and *abused* by our study definitional criteria) were categorized as *other* (not abusive) head trauma by their treating and consulting physicians at the time of hospital discharge.

DISCUSSION

The actual probability that a child's acute traumatic cranial injuries were inflicted is either zero or one. Unfortunately, in the absence of a gold standard, clinicians can rarely confirm or exclude AHT with complete certainty and are compelled instead to adopt a probabilistic approach to the diagnosis. To do so, clinicians formulate an initial (pretest) estimate of the probability of abuse, may elect to order additional evaluations for abuse, and use the results of those evaluations to formulate a revised (post-test) estimate of the probability of abuse.

Predictive Qualities of Isolated, Discriminating, and Reliable Variables

In the absence of patient-specific information, our results suggest that 0.45 (95 of 209) is a reasonable pretest estimate of abuse probability for an acutely head-injured infant or young child admitted for intensive care. Post-test probabilities of AHT for isolated, discriminating and reliable variables ranged from 0.1 to 0.86 (Fig. 1 and Table 5). Some variables, when positive, shifted probability of abuse upward greatly but changed it little when negative (see "Bruising of the ear(s), neck, or torso" in Fig. 1). Some variables, when positive, shifted probability of abuse downward greatly but changed it little when negative (see "Scalp, skull or epidural 'greatest depth' of visible [1° or 2°] brain injury" and "An isolated, unilateral, nondiastatic, linear, parietal skull fracture" in Fig. 1). Other variables were more useful when negative because negative findings largely excluded the diagnosis, whereas the positive finding increased probability only slightly (see "Any subdural hemorrhage(s) or fluid collection(s)" in Fig. 1). Some discriminating variables demonstrated poor inter-rater reliability (see "Mixed density subdural hemorrhage(s) or fluid collection(s)" in Table 4). A more complete understanding of the specific predictive qualities of isolated, discriminating, and reliable variables could improve screening accuracy.

Derivation of an AHT Clinical Prediction Rule

We did *not* attempt to derive a highly specific clinical prediction rule that could help future intensivists confirm AHT, if positive. Only 92 (53%) of 175 study patients designated

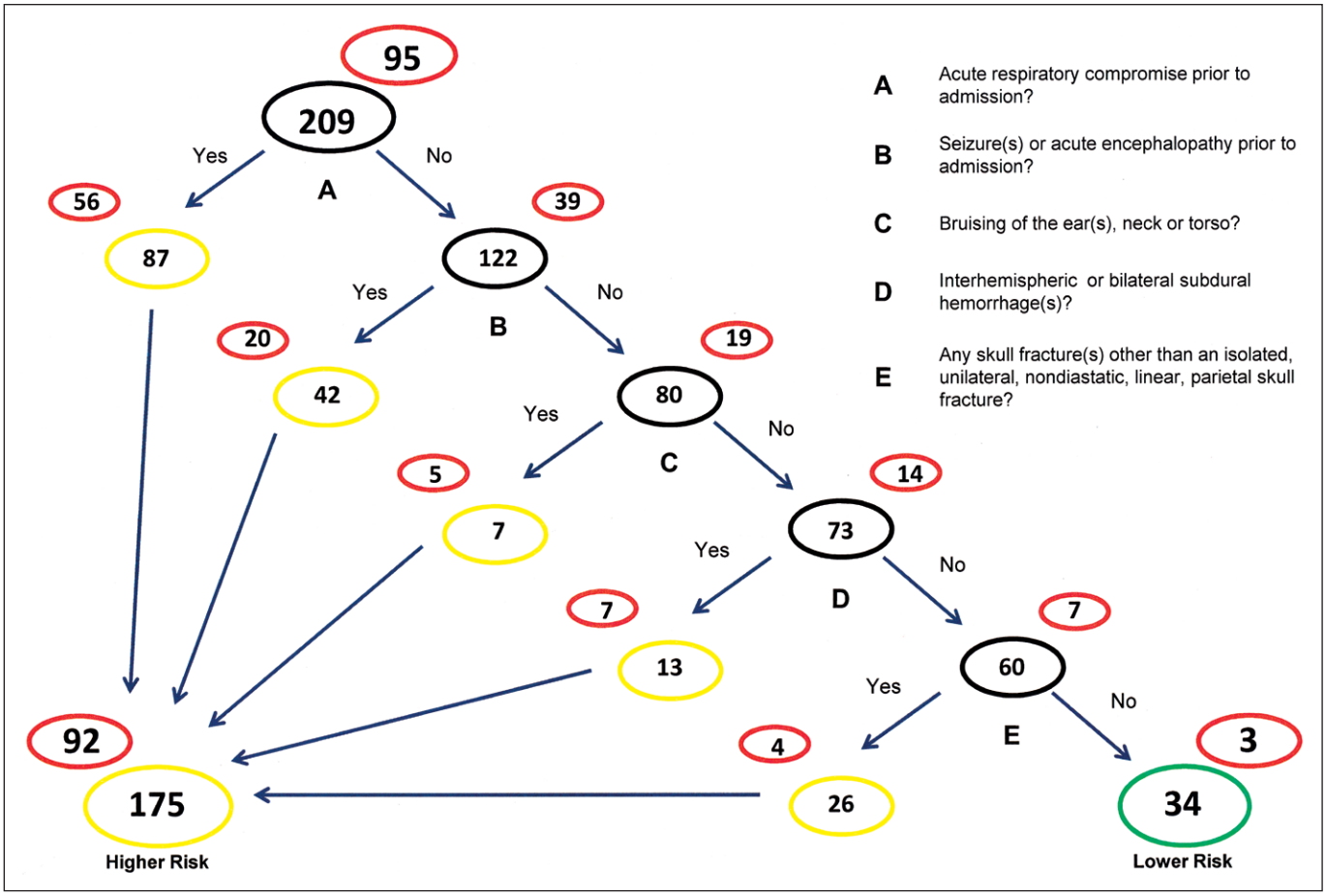

Figure 2. Recursive partitioning of 209 total and 95 abused study patients using five-variable abusive head trauma clinical prediction rule.

higher risk by our derived (five-variable) clinical prediction rule met study definitional criteria for AHT (Fig. 2). Even if validated, this clinical prediction rule should *not* be used to diagnose or confirm AHT.

We *did* attempt to derive a highly sensitive clinical prediction rule that could help future intensivists exclude AHT, if negative. To exclude abuse effectively, we hoped to derive a simple, reliable rule with 100% sensitivity and 100% negative predictive value. Our derived rule demonstrated sensitivity of 97%. Ninety-two (97%) of 95 patients that met study definitional criteria for abuse sorted as higher risk (Fig. 2). To achieve higher sensitivity, recursive partitioning created classification trees of unacceptable complexity that would limit their utility in a clinical setting. Negative predictive value was 91%. Thirty-one (91%) of 34 lower risk patients failed to meet study definitional criteria for abuse (Fig. 2).

Reasonable minds may differ in their interpretations of these results. Given the

TABLE 6. Classification Performance of our Derived (Five-Variable) Abusive Head Trauma Clinical Prediction Rule

		A Priori Study Definitional Criteria for Abusive Head Trauma	
		Abused	Other
Five-variable abusive head trauma clinical prediction rule	Higher risk	92	83
	Lower risk	3	31
		SENS (95% CI)	0.97 (0.90,0.99)
		SPEC (95% CI)	0.27 (0.20,0.37)
		PPV (95% CI)	0.53 (0.45,0.60)
		NPV (95% CI)	0.91 (0.75,0.98)
		LR(+)(95% CI)	1.33 (1.18,1.50)
		LR(−)(95% CI)	0.12 (0.04,0.37)

SENS = sensitivity; SPEC = specificity; PPV = positive predictive value; NPV = negative predictive value; LR(+) = positive likelihood ratio; and LR(−) = negative likelihood ratio.

3-Variable AHT Clinical Prediction Rule

- Acute respiratory compromise prior to hospital admission?
- Bruising of the ear(s), neck or torso?
- Interhemispheric or bilateral subdural hemorrhage(s) or fluid collection(s)?

	Study Definitional Criteria for AHT	
	Abused	*Other*
Higher Risk	87	46
Lower Risk	8	68
SENS	.92	
SPEC	.60	
PPV	.65	
NPV	.90	
LR(+)	2.27	
LR(-)	0.14	

4-Variable AHT Clinical Prediction Rule

- Acute respiratory compromise prior to admission?
- Seizure(s) or acute encephalopathy prior to admission?
- Bruising of the ear(s), neck or torso?
- Interhemispheric or bilateral subdural hemorrhage(s) or fluid collection(s)?

	Study Definitional Criteria for AHT	
	Abused	*Other*
Higher Risk	88	61
Lower Risk	7	53
SENS	.93	
SPEC	.47	
PPV	.59	
NPV	.88	
LR(+)	1.73	
LR(-)	0.16	

4-Variable AHT Clinical Prediction Rule

- Acute respiratory compromise prior to admission?
- Bruising of the ear(s), neck or torso?
- Interhemispheric or bilateral subdural hemorrhage(s) or fluid collection(s)?
- Any skull fracture(s) other than an isolated, unilateral, nondiastatic, linear, parietal skull fracture?

	Study Definitional Criteria for AHT	
	Abused	*Other*
Higher Risk	91	73
Lower Risk	4	41
SENS	.96	
SPEC	.36	
PPV	.56	
NPV	.91	
LR(+)	1.50	
LR(-)	0.12	

Figure 3. Classification performances of other abusive head trauma (AHT) clinical prediction rules with sensitivities >90%. SENS = sensitivity; SPEC = specificity; PPV = positive predictive value; NPV = negative predictive value; LR(+) = positive likelihood ratio; LR(-) = negative likelihood ratio.

danger of missing abuse, some will observe that no combination of variables perfectly excluded AHT and conclude that all acutely head-injured infants and young children admitted for intensive care (absent a motor vehicle collision) should have a thorough evaluation for abuse. In current practice, this is not happening. Because every abuse investigation is associated with risks to the child, family, and therapeutic relationship, some intensivists elect to forego a child abuse workup based on an individualized assessment of risk. Without data, these decisions could be subject to variability and bias (1–12). By facilitating calculation of a *patient-specific* estimate of abuse probability at or near the time of PICU admission, a validated, reliable, sensitive, AHT clinical prediction rule could provide a basis for discrimination that is more consistent and less subject to bias.

Although promising, our derived AHT clinical prediction rule is not yet ready for clinical application as a screening tool. We may be unable to validate its screening performance in a new patient population. An implementation study could demonstrate that clinicians are unable to apply or interpret a future, validated rule with sufficient reliability. Finally, research will be needed to verify that a future, validated, AHT clinical prediction rule improves the consistency of the medical response to severe child maltreatment or increases the diagnostic yield of abuse evaluations.

Study Strengths and Limitations

This study's primary strengths include its prospective, multicenter design, and methodology specifically designed and successfully executed to capture complete data, measure inter-rater reliability, and minimize sampling bias. Several variables with high likelihood ratios for abuse in our study patient population have been linked to abuse in other peer-reviewed studies (14, 15, 17, 18, 24–31). This is the first study to measure the specific predictive relationships between AHT and isolated clinical variables that are both discriminating *and* reliable (Fig. 1).

This study has limitations. These include 1) observational bias—in some cases, data re-

TABLE 7. Classification Performance of Study Definitional Criteria for Abusive Head Trauma Versus Treating (and Consulting) Physicians' Final, Consensus, Diagnostic Impressions at the Time of Hospital Discharge[a]

Study Definitional Criteria for Abusive Head Trauma	Treating Physicians' Final Forensic Impressions		
	Abused	Other	Totals
Abused	87	8	95
Other	16	98	114
Totals	103	106	209

[a]Two-tail McNemar's test = 0.152 (not significant).

garding predictor and outcome variables were not captured independently; 2) spectrum bias—very likely, our PICU patient population is poorly representative of the full clinical spectrum of pediatric AHT; 3) the Hawthorne effect—investigators' decisions to characterize cases as definitive or probable abuse may have been influenced by their active participation in this study; 4) design limitations—because our study was strictly observational, variations in the frequency, timing, and/or modalities of cranial imaging could have affected the validity of conclusions regarding specific injuries; 5) definitional limitations—there is no gold standard for AHT diagnosis; and 6) limited generalizability—future application of a validated AHT clinical prediction rule in non-PICU settings will require separate validation of screening performance in those locations.

Future Considerations

Broad implementation of a future, validated, AHT clinical prediction rule coupled with active participation in an AHT quality improvement collaborative could mitigate the effects of physician biases, improve the consistency of the medical response to severe child maltreatment, reduce unnecessary patient procedures and risks, and reduce missed cases.

CONCLUSIONS

Flawed decisions to launch (or to forego) an abuse evaluation can have serious adverse consequences for patients and their families. Such decisions are best made using a combination of intuitive and analytical reasoning. A more complete understanding of the specific predictive qualities of isolated, discriminating, and reliable variables could improve screening accuracy. If validated, a reliable, sensitive, AHT clinical prediction rule could be used by pediatric intensivists to calculate an evidence-based, patient-specific estimate of abuse probability that can inform—not dictate—their early decisions to launch (or to forego) an evaluation for abuse.

ACKNOWLEDGMENT

We thank the following persons for their ongoing guidance and support: Gautham Suresh, MD, Dartmouth-Hitchcock Medical Center, Lebanon, NH, and Daniel M. Lindberg, MD, Brigham and Women's Hospital, Boston, MA.

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